



COURSE CODE/TITLE:	CPE212 Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	8/25/26	DATE SUBMITTED:	8/26/26
NAME:	Aicon H. Keliste	INSTRUCTOR:	Engr. Jimlord Quejado

ACTIVITY NO. 5.1 **Implement Ansible roles in playbooks**

1. PROCEDURE OUTPUT

Task 1:

```
keliste@Keliste-Workstation: ~/CPE212_KELISTE$ ls
install_apache.yaml  README.md
keliste@Keliste-Workstation: ~/CPE212_KELISTE$ git pull
Already up to date.
keliste@Keliste-Workstation: ~/CPE212_KELISTE$
```



keliste@Keliste-Workstation: ~/CPE212_KELISTE

GNU nano 6.2

install_apache.yaml *

```
---
- hosts: all
  become: true
  tasks:

    - name: Update repository index
      apt:
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

    - name: install apache2 package
      apt:
        name: apache2
      when: ansible_distribution == "Ubuntu"

    - name: add PHP support for apache
      apt:
        name: libapache2-mod-php
      when: ansible_distribution == "Ubuntu"
```

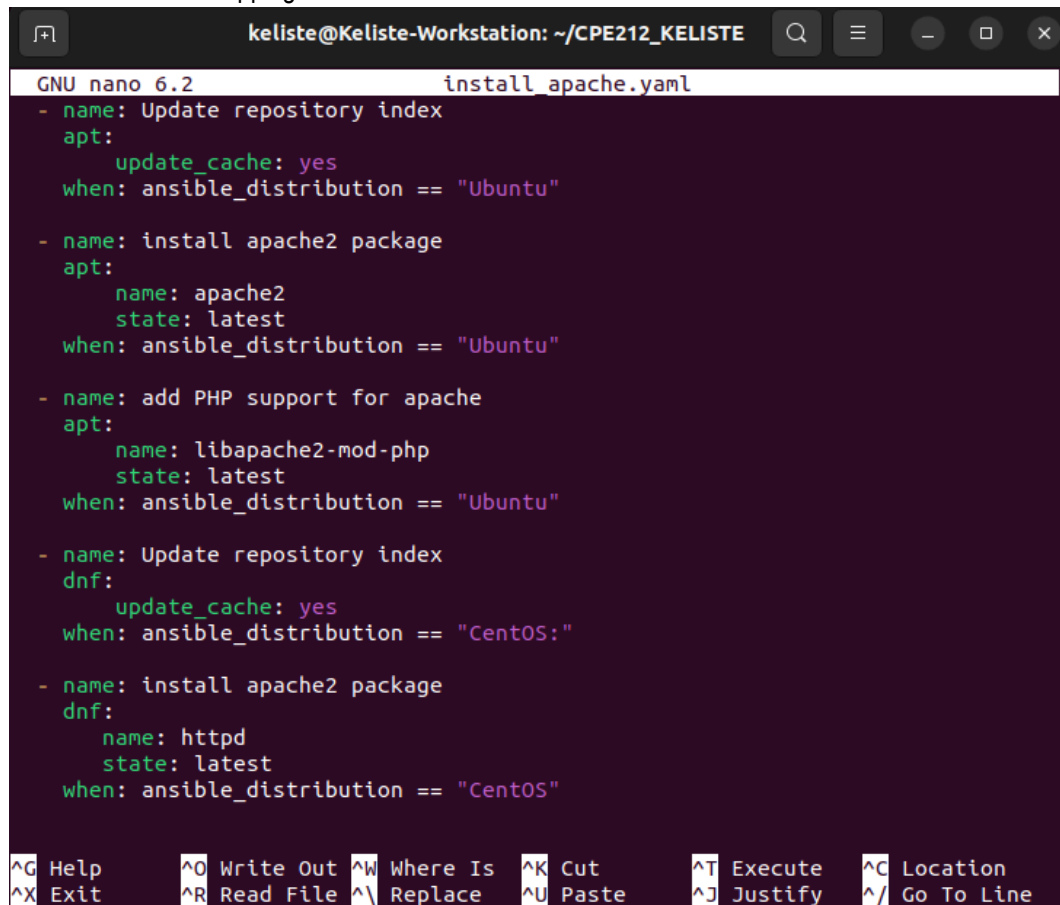
```
keliste@Keliste-Workstation:~/CPE212_KELISTE$ nano install_apache.yaml
keliste@Keliste-Workstation:~/CPE212_KELISTE$ ansible-playbook --ask-become-pass
install_apache.yaml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.104]
ok: [192.168.56.103]
[DEPRECATION WARNING]: Distribution centos 9 on host 192.168.56.105 should use
/usr/libexec/platform-python, but is using /usr/bin/python for backward
compatibility with prior Ansible releases. A future Ansible release will
default to using the discovered platform python for this host. See https://docs
.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for
more information. This feature will be removed in version 2.12. Deprecation
warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.105]

TASK [Update repository index] *****
skipping: [192.168.56.105]
```

Describe: It shows skipping because 192.168.56.105 is a CentOS machine



```
GNU nano 6.2 install_apache.yaml
- name: Update repository index
  apt:
    update_cache: yes
  when: ansible_distribution == "Ubuntu"

- name: install apache2 package
  apt:
    name: apache2
    state: latest
  when: ansible_distribution == "Ubuntu"

- name: add PHP support for apache
  apt:
    name: libapache2-mod-php
    state: latest
  when: ansible_distribution == "Ubuntu"

- name: Update repository index
  dnf:
    update_cache: yes
  when: ansible_distribution == "CentOS:"

- name: install apache2 package
  dnf:
    name: httpd
    state: latest
  when: ansible_distribution == "CentOS"
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
 ^X Exit ^R Read File ^_ Replace ^U Paste ^J Justify ^_ Go To Line

```
keliste@Keliste-Workstation:~/CPE212_KELISTE$ ansible-playbook --ask-become-pass
install_apache.yaml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.103]
[DEPRECATION WARNING]: Distribution centos 9 on host 192.168.56.105 should use
/usr/libexec/platform-python, but is using /usr/bin/python for backward
compatibility with prior Ansible releases. A future Ansible release will
default to using the discovered platform python for this host. See https://docs
.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for
more information. This feature will be removed in version 2.12. Deprecation
warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.105]
ok: [192.168.56.104]
```

```
keliste-centos@KelisteMain:~ — systemctl status httpd
keliste-centos@KelisteMain ~]$ systemctl status httpd
• httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disa
   Drop-In: /usr/lib/systemd/system/httpd.service.d
           └─php-fpm.conf
   Active: active (running) since Wed 2026-08-26 23:55:19 PST; 9min ago
   Docs: man:httpd.service(8)
   Main PID: 5703 (httpd)
   Status: "Total requests: 9; Idle/Busy workers 100/0;Requests/sec: 0.0154; >
   Tasks: 177 (limit: 10502)
   Memory: 21.8M (peak: 22.5M)
   CPU: 553ms
   CGroup: /system.slice/httpd.service
           └─5703 /usr/sbin/httpd -DFOREGROUND
             └─5710 /usr/sbin/httpd -DFOREGROUND
               └─5714 /usr/sbin/httpd -DFOREGROUND
                 └─5715 /usr/sbin/httpd -DFOREGROUND
                   └─5716 /usr/sbin/httpd -DFOREGROUND
lines 1-17/17 (END)
```

 **CentOS**

HTTP Server Test Page

This page is used to test the proper operation of the HTTP server after it has been installed. If you can read this page it means that this site is working properly. This server is powered by CentOS.

If you are a member of the general public:

The website you just visited is either experiencing problems or is undergoing routine maintenance.

If you would like to let the administrators of this website know that you've seen this page instead of the page you expected, you should send them e-mail. In general, mail sent to the name "webmaster" and directed to the website's domain should reach the appropriate person.

For example, if you experienced problems while visiting `www.example.com`, you should send e-mail to `"webmaster@example.com"`.

Important Note!

The CentOS Project has nothing to do with this website or its content,

Task 2:

```
GNU nano 6.2                                install_apache.yaml
--
- hosts: all
  become: true
  tasks:

  - name: Update repository index
    apt:
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: install apache2 and php packages for Ubuntu
    apt:
      name:
        - apache2
        - libapache2-mod-php
      state: latest
    when: ansible_distribution == "Ubuntu"

  - name: update repository index
    dnf:
      update_cache: yes
    when: ansible_distribution == "CentOS"

  - name: install apache2 and php packages for CentOS
    dnf:
      name:
        - httpd
        - php
      state: latest
    when: ansible_distribution == "CentOS"
```

[Read 30 lines]

```
keliste@Keliste-Workstation: ~/CPE212_KELISTE
keliste@Keliste-Workstation:~/CPE212_KELISTE$ nano install_apache.yaml
keliste@Keliste-Workstation:~/CPE212_KELISTE$ ansible-playbook --ask-become-pass
install_apache.yaml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.103]
ok: [192.168.56.104]
[DEPRECATION WARNING]: Distribution centos 9 on host 192.168.56.105 should use
/usr/libexec/platform-python, but is using /usr/bin/python for backward
compatibility with prior Ansible releases. A future Ansible release will
default to using the discovered platform python for this host. See https://docs
.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for
more information. This feature will be removed in version 2.12. Deprecation
warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.105]

TASK [Update repository index] *****
skipping: [192.168.56.105]
changed: [192.168.56.104]
changed: [192.168.56.103]

TASK [install apache2 and php packages for Ubuntu] *****
skipping: [192.168.56.105]
ok: [192.168.56.104]
ok: [192.168.56.103]

TASK [update repository index] *****
skipping: [192.168.56.103]
skipping: [192.168.56.104]
ok: [192.168.56.105]

TASK [install apache2 and php packages for CentOS] *****
skipping: [192.168.56.103]
skipping: [192.168.56.104]
ok: [192.168.56.105]

PLAY RECAP *****
192.168.56.103      : ok=3    changed=1    unreachable=0    failed=0    s
```

```
GNU nano 6.2      install_apache.yaml
---
- hosts: all
  become: true
  tasks:

  - name: install apache2 and php packages for Ubuntu
    apt:
      name:
        - apache2
        - libapache2-mod-php
      state: latest
      update_cache: yes
      when: ansible_distribution == "Ubuntu"

  - name: install apache2 and php packages for CentOS
    dnf:
      name:
        - httpd
        - php
      state: latest
      update_cache: yes
      when: ansible_distribution == "CentOS"
```



```

keliste@Keliste-Workstation: ~/CPE212_KELISTE
192.168.56.105 : ok=3 changed=0 unreachable=0 failed=0 s
kipped=2 rescued=0 ignored=0

keliste@Keliste-Workstation:~/CPE212_KELISTE$ nano install_apache.yaml
keliste@Keliste-Workstation:~/CPE212_KELISTE$ ansible-playbook --ask-become-pass
install_apache.yaml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.103]
ok: [192.168.56.104]
[DEPRECATION WARNING]: Distribution centos 9 on host 192.168.56.105 should use
/usr/libexec/platform-python, but is using /usr/bin/python for backward
compatibility with prior Ansible releases. A future Ansible release will
default to using the discovered platform python for this host. See https://docs
.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for
more information. This feature will be removed in version 2.12. Deprecation
warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.105]

TASK [install apache2 and php packages for Ubuntu] *****
skipping: [192.168.56.105]
ok: [192.168.56.104]
ok: [192.168.56.103]

TASK [install apache2 and php packages for CentOS] *****
skipping: [192.168.56.103]
skipping: [192.168.56.104]
ok: [192.168.56.105]

PLAY RECAP *****
192.168.56.103 : ok=2 changed=0 unreachable=0 failed=0 s
kipped=1 rescued=0 ignored=0
192.168.56.104 : ok=2 changed=0 unreachable=0 failed=0 s
kipped=1 rescued=0 ignored=0
192.168.56.105 : ok=2 changed=0 unreachable=0 failed=0 s
kipped=1 rescued=0 ignored=0

keliste@Keliste-Workstation:~/CPE212_KELISTE$

```

```

keliste@Keliste-Workstation: ~/CPE212_KELISTE
GNU nano 6.2 install_apache.yaml
---
- hosts: all
  become: true
  tasks:

  - name: install apache2 and php packages
    apt:
      name:
        - "{{ apache_package }}"
        - "{{ php_package }}"
      state: latest
      update_cache: yes

```



```
keliste@Keliste-Workstation: ~/CPE212_KELISTE
GNU nano 6.2 /etc/ansible/hosts
[myvirtualmachines]
192.168.56.103 apache_package=apache2 php_package=libapache2-mod-php
192.168.56.104 apache_package=apache2 php_package=libapache2-mod-php
192.168.56.105 ansible_user=keliste-centos apache_package=httpd php_package=php
```

```
GNU nano 6.2 install_apache.yaml
--
hosts: all
become: true
tasks:

- name: install apache2 and php packages
  package:
    name:
      - "{{ apache_package }}"
      - "{{ php_package }}"
    state: latest
    update_cache: yes
```

```
keliste@Keliste-Workstation:~/CPE212_KELISTE$ sudo nano /etc/ansible/hosts
[sudo] password for keliste:
keliste@Keliste-Workstation:~/CPE212_KELISTE$ nano install_apache.yaml
keliste@Keliste-Workstation:~/CPE212_KELISTE$ ansible-playbook --ask-become-pass in
stall_apache.yaml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.104]
ok: [192.168.56.103]
[DEPRECATION WARNING]: Distribution centos 9 on host 192.168.56.105 should use
/usr/libexec/platform-python, but is using /usr/bin/python for backward
compatibility with prior Ansible releases. A future Ansible release will default
to using the discovered platform python for this host. See https://docs.ansible.co
m/ansible/2.10/reference_appendices/interpreter_discovery.html for more
information. This feature will be removed in version 2.12. Deprecation warnings
can be disabled by setting deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.105]

TASK [install apache2 and php packages] *****
ok: [192.168.56.105]
ok: [192.168.56.104]
ok: [192.168.56.103]

PLAY RECAP *****
192.168.56.103      : ok=2    changed=0    unreachable=0    failed=0    skip
ped=0    rescued=0    ignored=0
192.168.56.104      : ok=2    changed=0    unreachable=0    failed=0    skip
ped=0    rescued=0    ignored=0
192.168.56.105      : ok=2    changed=0    unreachable=0    failed=0    skip
ped=0    rescued=0    ignored=0

keliste@Keliste-Workstation:~/CPE212_KELISTE$
```


2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. Create a playbook that could do the previous tasks in Red Hat OS.

```
[keliste-centos@KelisteMain ~]$ sudo dnf install epel-release -y
[sudo] password for keliste-centos:
Last metadata expiration check: 3:10:21 ago on Thu 27 Aug 2026 02:22:22 AM PST.
Dependencies resolved.
=====
Package Arch Version Repository Size
=====
Installing:
epel-release noarch 9-7.el9 extras-common 19 k
Installing weak dependencies:
epel-next-release noarch 9-7.el9 extras-common 8.1 k

Transaction Summary
=====
Install 2 Packages

Total download size: 27 k
Installed size: 29 k
Downloading Packages:
[MIRROR] epel-next-release-9-7.el9.noarch.rpm: Curl error (28): Timeout was reached for http://mirrors.tuna.tsinghua.edu.cn/centos-stream/SIGs/9-stream/extras/x86_64/extras-common/Packages/e/epel-next-release-9-7.el9.noarch.rpm [Failed to connect to mirrors.tuna.tsinghua.edu.cn port 80: Connection timed out]
[MIRROR] epel-release-9-7.el9.noarch.rpm: Curl error (28): Timeout was reached for http://mirrors.tuna.tsinghua.edu.cn/centos-stream/SIGs/9-stream/extras/x86_64/extras-common/Packages/e/epel-release-9-7.el9.noarch.rpm [Failed to connect to mirrors.tuna.tsinghua.edu.cn port 80: Connection timed out]

[keliste-centos@KelisteMain ~]$ sudo dnf install ansible -y
Extra Packages for Enterprise100% [=====] 547 kB/s | 6.2 MB 00:00 ETA

[keliste-centos@KelisteMain ~]$ sudo dnf install git -y
[sudo] password for keliste-centos:
Last metadata expiration check: 0:12:06 ago on Thu 27 Aug 2026 05:35:40 AM PST.
Dependencies resolved.
=====
Package Architecture Version Repository Size
=====
Installing:
git x86_64 2.52.0-1.el9 appstream 39 k
Installing dependencies:
git-core-doc noarch 2.52.0-1.el9 appstream 3.1 M
perl-Error noarch 1:0.17029-7.el9 appstream 42 k
perl-Git noarch 2.52.0-1.el9 appstream 37 k
perl-TermReadKey x86_64 2.38-11.el9 appstream 37 k
perl-lib x86_64 0.65-483.el9 appstream 15 k

Transaction Summary
=====
Install 6 Packages

Total download size: 3.3 M
Installed size: 19 M
Downloading Packages:
(1/6): git-2.52.0-1.el9.x86_64.rpm 71 kB/s | 39 kB 00:00
(2/6): perl-Error-0.17029-7.el9.noarch.rpm 77 kB/s | 42 kB 00:00
(3/6): perl-TermReadKey-2.38-11.el9.x86_64.rpm 174 kB/s | 37 kB 00:00
(4/6): perl-lib-0.65-483.el9.x86_64.rpm 91 kB/s | 15 kB 00:00
```

```
[keliste-centos@KelisteMain ~]$ git clone https://github.com/AiconKeliste/CPE212_KELISTE.git
Cloning into 'CPE212_KELISTE'...
remote: Enumerating objects: 18, done.
remote: Counting objects: 100% (18/18), done.
remote: Compressing objects: 100% (14/14), done.
remote: Total 18 (delta 0), reused 12 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (18/18), done.
[keliste-centos@KelisteMain ~]$ cd CPE212_KELISTE
[keliste-centos@KelisteMain CPE212_KELISTE]$ ls
hosts  install_apache.yaml  README.md
[keliste-centos@KelisteMain CPE212_KELISTE]$
```

```
[keliste-centos@KelisteMain CPE212_KELISTE]$ ssh-keygen -t rsa
Generating public/private rsa key pair.
Enter file in which to save the key (/home/keliste-centos/.ssh/id_rsa):
Enter passphrase for "/home/keliste-centos/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/keliste-centos/.ssh/id_rsa
Your public key has been saved in /home/keliste-centos/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:CI8AFb6P+4hMoR1G189vZqlobHan/pGP4dKY9E50T68 keliste-centos@KelisteMain
The key's randomart image is:
+---[RSA 3072]-----+
|..O.                |
| O .                |
| + +                |
| . + * .            |
| .+ . = S           |
|.O.+ .. O           |
|..O... =% .          |
|O. O.=X+B .         |
|..O=oo+** E..       |
+---[SHA256]-----+
```

```
[keliste-centos@KelisteMain CPE212_KELISTE]$ ssh-copy-id keliste@192.168.56.103
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: ssh-add -L
The authenticity of host '192.168.56.103 (192.168.56.103)' can't be established.
ED25519 key fingerprint is SHA256:L8+xoAX1tr2m6AWwQnQDNP/E/x841EXcASYshAM2+rE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already
installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install t
he new keys
keliste@192.168.56.103's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'keliste@192.168.56.103'"
and check to make sure that only the key(s) you wanted were added.

[keliste-centos@KelisteMain CPE212_KELISTE]$ ssh keliste@192.168.56.103
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.
```



```
[keliste-centos@KelisteMain CPE212_KELISTE]$ ssh-copy-id keliste@192.168.56.104
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: ssh-add -L
The authenticity of host '192.168.56.104 (192.168.56.104)' can't be established.
ED25519 key fingerprint is SHA256:FiQxJc8PnOnmuV0eQBWd+/GDKdcVCeLNUjWkaRdlza8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already
installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install t
he new keys
keliste@192.168.56.104's password:
Permission denied, please try again.
keliste@192.168.56.104's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'keliste@192.168.56.104'"
and check to make sure that only the key(s) you wanted were added.

[keliste-centos@KelisteMain CPE212_KELISTE]$ ssh keliste@192.168.56.104
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

connection to 192.168.56.104 closed.
[keliste-centos@KelisteMain CPE212_KELISTE]$ ssh-copy-id keliste@192.168.56.102
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: ssh-add -L
The authenticity of host '192.168.56.102 (192.168.56.102)' can't be established.
ED25519 key fingerprint is SHA256:TPundoXtcUkDI/hKP/gGS6aFmYgyMFdaSiozKRUC/YI.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already
installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install t
he new keys
keliste@192.168.56.102's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'keliste@192.168.56.102'"
and check to make sure that only the key(s) you wanted were added.

[keliste-centos@KelisteMain CPE212_KELISTE]$ ssh keliste@192.168.56.102
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.
```

```
[keliste-centos@KelisteMain CPE212_KELISTE]$ ansible-playbook -i hosts --ask-become-pass install_apache.y
aml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.104]
ok: [192.168.56.105]
ok: [192.168.56.102]
ok: [192.168.56.103]

TASK [install apache2 and php packages] *****
fatal: [192.168.56.102]: FAILED! => {"msg": "The task includes an option with an undefined variable. The
error was: 'apache_package' is undefined. 'apache_package' is undefined\n\nThe error appears to be in '/h
ome/keliste-centos/CPE212_KELISTE/install_apache.yaml': line 6, column 5, but may\nbe elsewhere in the fi
le depending on the exact syntax problem.\n\nThe offending line appears to be:\n\n\n - name: install apa
che2 and php packages\n    ^ here\n"}
ok: [192.168.56.105]
```

```
GNU nano 5.6.1 hosts Modified
[myvirtualmachines]
192.168.56.102 ansible_user=kellite apache_package=apache2 php_package=libapache2-mod-php
192.168.56.103 ansible_user=kellite apache_package=apache2 php_package=libapache2-mod-php
192.168.56.104 ansible_user=kellite apache_package=apache2 php_package=libapache2-mod-php
192.168.56.105 ansible_user=kellite-centos apache_package=httpd php_package=php

[keliste-centos@KelisteMain CPE212_KELISTE]$ ansible-playbook -i hosts --ask-become-pass install_apache.yaml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.104]
ok: [192.168.56.103]
ok: [192.168.56.102]
ok: [192.168.56.105]

TASK [install apache2 and php packages] *****
ok: [192.168.56.104]
ok: [192.168.56.103]
ok: [192.168.56.105]
changed: [192.168.56.102]

PLAY RECAP *****
192.168.56.102      : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
192.168.56.103      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
192.168.56.104      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
192.168.56.105      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

[keliste-centos@KelisteMain CPE212_KELISTE]$
```

Reflections:

Answer the following:

- Why do you think refactoring of playbook codes is important?
 - Refactoring makes the playbook easier to read and maintain, since one variable-based task replaces multiple OS-specific ones. It also cuts execution time, since Ansible runs fewer tasks per host.
- When do we use the “when” command in playbook?
 - I use when whenever a task should only run on hosts meeting a specific condition, like `ansible_distribution`. It was essential once I had a mixed Ubuntu/CentOS environment, since each needed different package managers. Without it, Ansible would try running apt on CentOS and fail instead of skipping the task.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Claude AI**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Command Assistance**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I asked claude ai on how to transfer the directory from ubuntu into centos vm via
ssh and git.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.